

Nmap Hands-On Activities

Six lab exercises for teaching network scanning and reconnaissance with Nmap on Kali Linux

These activities give students hands-on practice with Nmap, progressing from simple network discovery to scripted vulnerability enumeration and a capture-the-flag style challenge. Each uses safe, legal, purpose-built targets so students can practice freely without any legal risk.

1. Home Network Discovery Sweep

Students scan their own home network (or a lab VLAN) to build a live inventory of connected devices.

- Run `nmap -sn 192.168.1.0/24` to find every live device on their network
- Identify each device (router, phone, laptop, smart TV, etc.) and document its IP address and MAC vendor

Deliverable: A simple network diagram or table listing every discovered host.

IP Address	MAC Address	MAC Vendor	Device Identification
192.168.1.1	B0:8B:92:B7:97:FA	ZTE	Main Fiber Router / Gateway (gpon.net)
192.168.1.2	A8:29:48:4A:9B:D5	TP-Link Systems	Network Switch / Access Point
192.168.1.3	98:BA:5F:A8:E7:3D	TP-Link Systems	Network Switch / Access Point
192.168.1.4	A8:6E:84:05:14:CC	TP-Link Systems	Network Switch / Access Point
192.168.1.5	A8:6E:84:05:1A:22	TP-Link Systems	Network Switch / Access Point
192.168.1.6	A8:6E:84:05:14:C8	TP-Link Systems	Network Switch / Access Point
192.168.1.7	0C:EF:15:A8:CC:63	TP-Link Systems	Network Switch / Extender
192.168.1.9	0C:EF:15:A8:D2:48	TP-Link Systems	Network Switch / Extender
192.168.1.11	0C:EF:15:A8:CC:63	TP-Link Systems	Network Switch / Extender
192.168.1.13	4E:73:76:4A:AE:EF	Unknown	Mobile Device (Randomized MAC)
192.168.1.14	46:F8:CC:C5:22:6A	Unknown	Mobile Device (Randomized MAC)
192.168.1.15	42:82:80:98:15:F5	Unknown	Mobile Device (Randomized MAC)
192.168.1.16	6E:30:54:36:1C:11	Unknown	Mobile Device (Randomized MAC)
192.168.1.18	D2:EC:8A:1D:44:97	Unknown	Mobile Device (Randomized MAC)
192.168.1.19	62:9A:33:84:F8:DC	Unknown	Mobile Device (Randomized MAC)
192.168.1.21	3C:84:6A:65:99:8D	TP-Link Technologies	Smart Home Device / Plug
192.168.1.22	0C:EF:15:A8:C5:F1	TP-Link Systems	Network Switch / Extender
192.168.1.23	F4:3B:D8:7C:B2:9E	Intel Corporate	Windows/Linux Laptop or Desktop
192.168.1.26	70:08:10:0A:FD:ED	Intel Corporate	Windows/Linux Laptop or Desktop
192.168.1.28	B0:47:E9:BD:20:4A	Intel Corporate	Windows/Linux Laptop or Desktop
192.168.1.29	B2:B6:A8:7A:C5:3C	Unknown	Mobile Device (Randomized MAC)
192.168.1.31	0C:EF:15:A8:D4:F5	TP-Link Systems	Network Switch / Extender

192.168.1.32	0E:D1:17:6B:EA:2F	Unknown	Mobile Device (Randomized MAC)
192.168.1.37	0C:EF:15:A8:BD:6E	TP-Link Systems	Network Switch / Extender
192.168.1.40	4A:00:CB:F2:18:3D	Unknown	Mobile Device (Randomized MAC)
192.168.1.43	62:68:FA:65:A8:09	Unknown	Mobile Device (Randomized MAC)
192.168.1.44	FE:85:1C:E5:D8:14	Unknown	Mobile Device (Randomized MAC)
192.168.1.49	8E:B3:AE:B0:D2:94	Unknown	Mobile Device (Randomized MAC)
192.168.1.51	40:C7:3C:0D:02:BE	Intel Corporate	Windows/Linux Laptop or Desktop
192.168.1.53	70:08:10:0A:FD:DE	Intel Corporate	Windows/Linux Laptop or Desktop
192.168.1.57	40:C7:3C:0C:F5:17	Intel Corporate	Windows/Linux Laptop or Desktop
192.168.1.58	EC:8E:77:C1:71:36	Intel Corporate	Windows/Linux Laptop or Desktop
192.168.1.60	40:C7:3C:0A:F7:5D	Intel Corporate	Windows/Linux Laptop or Desktop
192.168.1.61	40:C7:3C:0A:F7:99	Intel Corporate	Windows/Linux Laptop or Desktop
192.168.1.64	24:EB:16:F8:B0:CE	Intel Corporate	Windows/Linux Laptop or Desktop
192.168.1.72	40:C7:3C:0A:F7:03	Intel Corporate	Windows/Linux Laptop or Desktop
192.168.1.78	0C:EF:15:A8:C1:39	TP-Link Systems	Network Switch / Extender
192.168.1.80	EC:8E:77:BE:FD:84	Intel Corporate	Windows/Linux Laptop or Desktop
192.168.1.81	58:41:46:4B:05:EC	Guangzhou Shiyuan Tech	Smart TV / Interactive Board
192.168.1.88	70:08:10:0A:F7:FD	Intel Corporate	Windows/Linux Laptop or Desktop
192.168.1.94	0C:EF:15:A8:C7:F7	TP-Link Systems	Network Switch / Extender
192.168.1.96	6E:8E:41:76:31:EB	Unknown	Mobile Device (Randomized MAC)
192.168.1.97	70:68:71:E2:6E:75	FN-Link Technology	IoT Module / Smart Appliance
192.168.1.98	0C:EF:15:A8:BA:B1	TP-Link Systems	Network Switch / Extender
192.168.1.102	76:31:A3:86:D5:F7	Unknown	Mobile Device (Randomized MAC)
192.168.1.103	CE:DF:CD:70:C1:DA	Unknown	Mobile Device (Randomized MAC)
192.168.1.105	E2:05:DD:7A:9C:CD	Unknown	Mobile Device (Randomized MAC)
192.168.1.106	60:C7:BE:A8:D9:3B	Realme Mobile	Android Smartphone
192.168.1.107	0C:EF:15:A8:D9:4C	TP-Link Systems	Network Switch / Extender
192.168.1.108	24:EE:9A:E7:85:0B	Intel Corporate	Windows/Linux Laptop or Desktop
192.168.1.109	Local Interface	N/A	Kali Linux VM (Scanning Host)
192.168.1.110	C2:1B:C6:3E:EE:24	Unknown	Mobile Device (Randomized MAC)
192.168.1.112	06:BB:24:BE:A9:EE	Unknown	Mobile Device (Randomized MAC)
192.168.1.113	32:BA:D4:F4:FD:98	Unknown	Mobile Device (Randomized MAC)

192.168.1.114	40:C7:3C:0A:F7:08	Intel Corporate	Windows/Linux Laptop or Desktop
192.168.1.115	5A:A4:FB:19:21:8F	Unknown	Mobile Device (Randomized MAC)
192.168.1.118	40:C7:3C:0D:02:D7	Intel Corporate	Windows/Linux Laptop or Desktop
192.168.1.120	BE:C2:AB:82:21:F6	Unknown	Mobile Device (Randomized MAC)
192.168.1.121	F4:8C:50:34:D6:D6	Intel Corporate	Windows/Linux Laptop or Desktop
192.168.1.123	D6:C9:D9:FD:A0:41	Unknown	Mobile Device (Randomized MAC)
192.168.1.124	0C:EF:15:A8:D9:4C	TP-Link Systems	Network Switch / Extender
192.168.1.126	0C:EF:15:A8:C1:39	TP-Link Systems	Network Switch / Extender

2. Metasploitable 2 Port & Service Enumeration

Using the intentionally vulnerable Metasploitable 2 VM (free download, isolated VM only).

- Run **nmap -sV -p- <target>** against the VM
- List every open port, the service, and the version Nmap detected
- Research one CVE tied to an outdated service they found (e.g. vsftpd 2.3.4)

Deliverable: A short enumeration report mapping ports → services → potential risk.

Target Host: 192.168.56.101

Scanning Host: 192.168.56.102 (Kali Linux)

Command Executed: nmap -sV -p- 192.168.56.101

Assessment Type: Internal Vulnerability & Service Enumeration

Enumeration Mapping Table (Ports → Services → Risk)

The left screenshot shows the output of an nmap scan for 192.168.56.101. The scan results are as follows:

```

Nmap scan report for 192.168.56.101
Host is up (0.00028s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 5ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rshcd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  shell        Metakit rshd
1099/tcp  open  java-rmi     GNU Classpath gmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
3632/tcp  open  distccd      distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
6697/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
8787/tcp  open  drb          Ruby DRB RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drbb)
34989/tcp open  nlockmgr     1-4 (RPC #100021)
37502/tcp open  status       1 (RPC #100024)
51619/tcp open  mountd       1-3 (RPC #100005)
57504/tcp open  java-rmi     GNU Classpath gmiregistry
MAC Address: 08:00:27:0D:4F:D8 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OS: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 139.45 seconds
  
```

The right screenshot shows the output of an ifconfig command on the Metasploitable2 VM. The output is as follows:

```

msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:0d:4f:d8
          inet addr:192.168.56.101  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fedd:4fd0/64  Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:2 errors:0 dropped:0 overruns:0 frame:0
          TX packets:25 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1180 (1.1 KB)  TX bytes:3630 (3.5 KB)
          Base address:0xd020  Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128  Scope:Host
          UP LOOPBACK RUNNING  MTU:16384  Metric:1
          RX packets:96 errors:0 dropped:0 overruns:0 frame:0
          TX packets:96 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:21437 (20.9 KB)  TX bytes:21437 (20.9 KB)

msfadmin@metasploitable:~$
msfadmin@metasploitable:~$
msfadmin@metasploitable:~$
  
```

3. Scan Type Comparison Lab

Students compare scan techniques on the same target and reason about the trade-offs.

- Run -sS, -sT, and -sU against the same host and time each with time nmap ...
- Capture the traffic in Wireshark alongside each scan

Deliverable: A short write-up comparing the speed, packet behavior, and stealth of each scan type.

```
Session Actions Edit View Help
(kali㉿kali)-[~]
$ sudo nmap -sS 192.168.1.1
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-24 02:43 -0400
Nmap scan report for gpon.net (192.168.1.1)
Host is up (0.047s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE      SERVICE
23/tcp    filtered  telnet
53/tcp    open       domain
80/tcp    open       http
443/tcp   open       https
MAC Address: B0:8B:92:B7:97:FA (zte)

Nmap done: 1 IP address (1 host up) scanned in 1.67 seconds

(kali㉿kali)-[~]
$ nmap -sT 192.168.1.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-24 02:48 -0400
Nmap scan report for gpon.net (192.168.1.1)
Host is up (0.026s latency).
Not shown: 996 closed tcp ports (conn-refused)
PORT      STATE      SERVICE
23/tcp    filtered  telnet
53/tcp    open       domain
80/tcp    open       http
443/tcp   open       https
MAC Address: B0:8B:92:B7:97:FA (zte)

Nmap done: 1 IP address (1 host up) scanned in 15.29 seconds

(kali㉿kali)-[~]
$ sudo nmap -sU 192.168.1.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-24 02:50 -0400
Nmap scan report for gpon.net (192.168.1.1)
Host is up (0.0089s latency).
Not shown: 998 closed udp ports (port-unreach)
PORT      STATE      SERVICE
53/udp    open       domain
67/udp    open|filtered dhcp
MAC Address: B0:8B:92:B7:97:FA (zte)

Nmap done: 1 IP address (1 host up) scanned in 117.28 seconds

(kali㉿kali)-[~]
$
```

Kali Linux 2026.2 virtualbox-amd64 [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

*eth0

ip.addr == 192.168.1.1

No.	Time	Source	Destination	Protocol	Length	Info
249	55.461528292	192.168.1.109	192.168.1.1	TCP	58	36789 → 993 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
250	55.461670206	192.168.1.109	192.168.1.1	TCP	58	36789 → 1723 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
251	55.461814562	192.168.1.109	192.168.1.1	TCP	58	36789 → 445 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
252	55.461965490	192.168.1.109	192.168.1.1	TCP	58	36789 → 80 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
253	55.462094378	192.168.1.109	192.168.1.1	TCP	58	36789 → 554 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
254	55.462225599	192.168.1.109	192.168.1.1	TCP	58	36789 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
255	55.462363746	192.168.1.109	192.168.1.1	TCP	58	36789 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
256	55.473264044	192.168.1.1	192.168.1.109	TCP	60	8880 → 36789 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
257	55.473264065	192.168.1.1	192.168.1.109	TCP	60	22 → 36789 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
258	55.473264086	192.168.1.1	192.168.1.109	TCP	60	8888 → 36789 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
261	55.473264106	192.168.1.1	192.168.1.109	TCP	60	993 → 36789 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
262	55.473264127	192.168.1.1	192.168.1.109	TCP	60	1723 → 36789 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
263	55.473264147	192.168.1.1	192.168.1.109	TCP	60	445 → 36789 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
264	55.473307548	192.168.1.1	192.168.1.109	TCP	60	80 → 36789 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1460
265	55.473307643	192.168.1.1	192.168.1.109	TCP	60	554 → 36789 [SYN, ACK] Seq=1 Ack=1 Win=0 Len=0
266	55.473307672	192.168.1.1	192.168.1.109	TCP	60	443 → 36789 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1460
267	55.473310005	192.168.1.109	192.168.1.1	TCP	54	36789 → 60 [RST] Seq=1 Win=0 Len=0
268	55.473372929	192.168.1.109	192.168.1.1	TCP	54	36789 → 443 [RST] Seq=1 Win=0 Len=0
269	55.475741164	192.168.1.109	192.168.1.1	TCP	58	36789 → 53 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
270	55.475911168	192.168.1.109	192.168.1.1	TCP	58	36789 → 111 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
271	55.476046658	192.168.1.109	192.168.1.1	TCP	58	36789 → 1025 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
272	55.476175642	192.168.1.109	192.168.1.1	TCP	58	36789 → 113 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
273	55.476306539	192.168.1.109	192.168.1.1	TCP	58	36789 → 21 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
274	55.476432659	192.168.1.109	192.168.1.1	TCP	58	36789 → 139 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
275	55.476557844	192.168.1.109	192.168.1.1	TCP	58	36789 → 110 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
276	55.476684197	192.168.1.109	192.168.1.1	TCP	58	36789 → 25 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
277	55.476970778	192.168.1.109	192.168.1.1	TCP	58	36789 → 113 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
278	55.477130870	192.168.1.109	192.168.1.1	TCP	58	36789 → 199 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
279	55.477258068	192.168.1.109	192.168.1.1	TCP	58	36789 → 995 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
280	55.477380939	192.168.1.109	192.168.1.1	TCP	58	36789 → 135 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
281	55.477562433	192.168.1.109	192.168.1.1	TCP	58	36789 → 3396 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
282	55.478181308	192.168.1.109	192.168.1.1	TCP	58	36789 → 1720 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
283	55.478419844	192.168.1.109	192.168.1.1	TCP	58	36789 → 587 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
284	55.478958887	192.168.1.109	192.168.1.1	TCP	58	36789 → 3389 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
285	55.479607515	192.168.1.109	192.168.1.1	TCP	58	36789 → 256 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
286	55.479735088	192.168.1.109	192.168.1.1	TCP	58	36789 → 5900 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
287	55.480090305	192.168.1.109	192.168.1.1	TCP	60	511 → 36789 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
288	55.481095908	192.168.1.1	192.168.1.109	TCP	60	53 → 36789 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1460

Frame 251: Packet, 58 bytes on wire (464 bits), 58 bytes captured (464 bits) on interface eth0, Ethernet II, Src: PCSSysntec, Sa:87:bc (08:00:27:5a:87:bc), Dst: zte.b7:97:fa (b8:0b:92:b7:97:fa), Internet Protocol Version 4, Src: 192.168.1.109, Dst: 192.168.1.1, Transmission Control Protocol, Src Port: 36789, Dst Port: 445, Seq: 0, Len: 0

wireshark_eth0p4UAW3.pcapng

Packets: 2412 · Displayed: 2003 (83.0%) · Dropped: 0 (0.0%) · Profile: Default

Kali Linux 2026.2 virtualbox-amd64 [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

*eth0

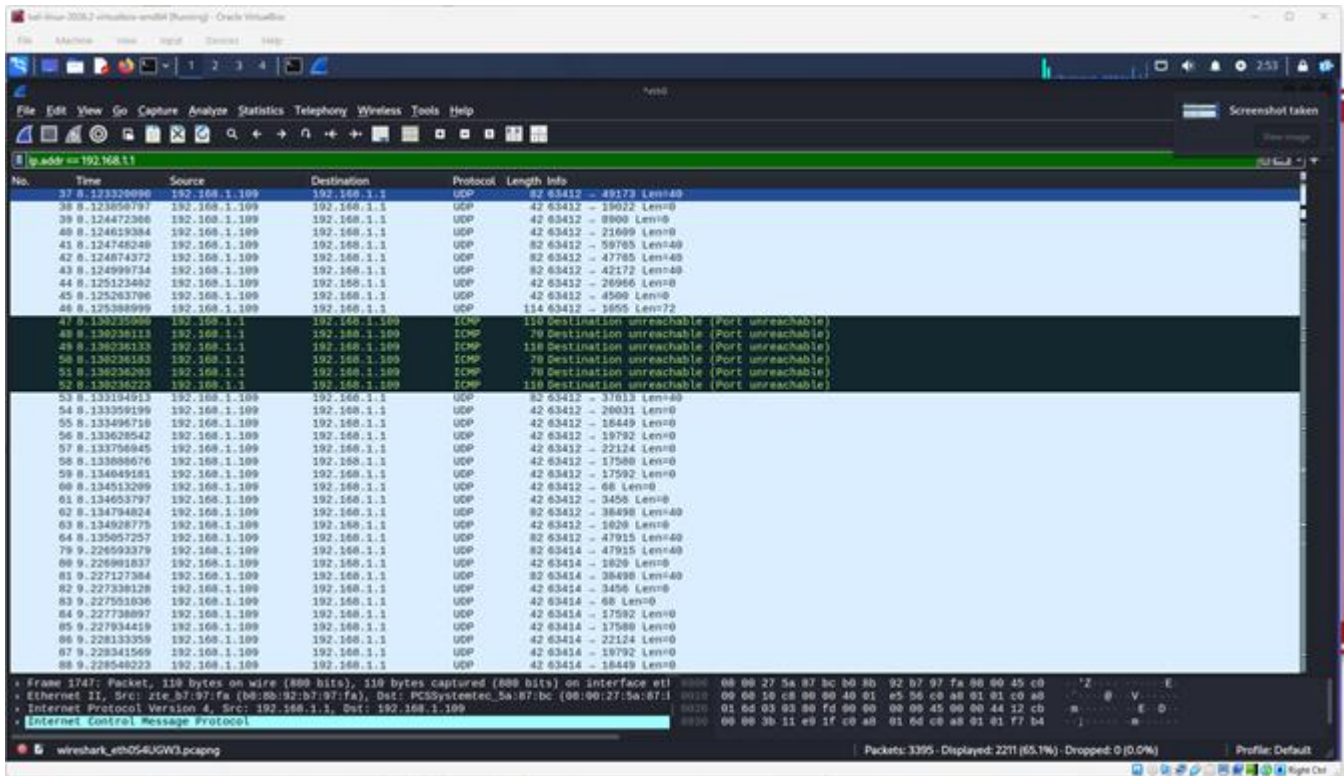
ip.addr == 192.168.1.1

No.	Time	Source	Destination	Protocol	Length	Info
2515	29.384329448	192.168.1.109	192.168.1.1	TCP	74	58454 → 4005 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=18849622987 TSecr=0 WS=256
2516	29.390343963	192.168.1.109	192.168.1.1	TCP	74	33884 → 3333 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1886435849 TSecr=0 WS=256
2518	29.409356178	192.168.1.109	192.168.1.1	TCP	74	40844 → 12000 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1205610417 TSecr=0 WS=256
2519	29.415548398	192.168.1.109	192.168.1.1	TCP	74	36020 → 3208 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1887479882 TSecr=0 WS=256
2520	29.425303178	192.168.1.109	192.168.1.1	TCP	74	35378 → 1005 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1486752073 TSecr=0 WS=256
2521	29.480526245	192.168.1.109	192.168.1.1	TCP	74	58450 → 4005 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1318993348 TSecr=0 WS=256
2522	29.490665681	192.168.1.109	192.168.1.1	TCP	74	33888 → 3333 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=12913603725 TSecr=0 WS=256
2523	29.506871149	192.168.1.109	192.168.1.1	TCP	74	46950 → 12000 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=12016880493 TSecr=0 WS=256
2526	29.512105055	192.168.1.1	192.168.1.109	TCP	60	3333 → 33884 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2527	29.512105411	192.168.1.1	192.168.1.109	TCP	60	4005 → 58454 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2528	29.512105521	192.168.1.1	192.168.1.109	TCP	60	1005 → 35378 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2529	29.512105604	192.168.1.1	192.168.1.109	TCP	60	12000 → 40844 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2530	29.512105649	192.168.1.1	192.168.1.109	TCP	60	3208 → 36020 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2531	29.512105732	192.168.1.1	192.168.1.109	TCP	60	4005 → 58450 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2532	29.515742775	192.168.1.1	192.168.1.109	TCP	60	3333 → 33888 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2533	29.515743017	192.168.1.1	192.168.1.109	TCP	60	12000 → 46950 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2534	29.517581876	192.168.1.109	192.168.1.1	TCP	74	32084 → 8291 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=12763648662 TSecr=0 WS=256
2535	29.519002687	192.168.1.109	192.168.1.1	TCP	74	53214 → 40175 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=13186558453 TSecr=0 WS=256
2536	29.531890926	192.168.1.1	192.168.1.109	TCP	60	8291 → 52084 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2537	29.530377833	192.168.1.109	192.168.1.1	TCP	74	45042 → 5440 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=12404493243 TSecr=0 WS=256
2538	29.530531442	192.168.1.1	192.168.1.109	TCP	60	49175 → 55234 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2539	29.540526048	192.168.1.109	192.168.1.1	TCP	74	54322 → 4005 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=12175829908 TSecr=0 WS=256
2540	29.550374155	192.168.1.109	192.168.1.1	TCP	60	5440 → 53042 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2541	29.550552741	192.168.1.109	192.168.1.1	TCP	74	40444 → 1020 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=12801638127 TSecr=0 WS=256
2542	29.556539484	192.168.1.1	192.168.1.109	TCP	60	8005 → 54022 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2543	29.560524531	192.168.1.109	192.168.1.1	TCP	74	33874 → 2875 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=12625772252 TSecr=0 WS=256
2544	29.570276137	192.168.1.109	192.168.1.1	TCP	74	58248 → 9610 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=11737618504 TSecr=0 WS=256
2545	29.581323211	192.168.1.109	192.168.1.1	TCP	60	1020 → 40444 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2546	29.583324882	192.168.1.1	192.168.1.109	TCP	60	2875 → 33874 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2547	29.583324887	192.168.1.1	192.168.1.109	TCP	60	9610 → 58248 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2548	29.590243968	192.168.1.109	192.168.1.1	TCP	74	58454 → 1272 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=14195861030 TSecr=0 WS=256
2549	29.600258434	192.168.1.109	192.168.1.1	TCP	74	55130 → 8182 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=14871445399 TSecr=0 WS=256
2550	29.604020203	192.168.1.109	192.168.1.1	TCP	60	8182 → 55130 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2551	29.610250158	192.168.1.109	192.168.1.1	TCP	74	30510 → 2306 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=12064388526 TSecr=0 WS=256
2552	29.618050390	192.168.1.1	192.168.1.109	TCP	60	6102 → 55130 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2553	29.618263747	192.168.1.1	192.168.1.109	TCP	60	2306 → 30510 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2554	29.620420115	192.168.1.109	192.168.1.1	TCP	74	52741 → 8291 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=12115072210 TSecr=0 WS=256
2555	29.705334578	192.168.1.109	192.168.1.1	TCP	60	8291 → 52741 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0

Frame 82: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface eth0, Ethernet II, Src: PCSSysntec, Sa:87:bc (08:00:27:5a:87:bc), Dst: zte.b7:97:fa (b8:0b:92:b7:97:fa), Internet Protocol Version 4, Src: 192.168.1.109, Dst: 192.168.1.1, Transmission Control Protocol, Src Port: 59912, Dst Port: 23, Seq: 0, Len: 0

eth0: <live capture in progress>

Packets: 2711 · Displayed: 2370 (87.4%) · Profile: Default



1. Empirical Results & Performance Comparison

Scan Type	Nmap Command	Execution Time	Open Ports Identified	Wireshark Packet Signature
TCP SYN (Stealth)	sudo nmap -sS 192.168.1.1	1.67 seconds	53/tcp, 80/tcp, 443/tcp	SYN $\rightarrow$ SYN/ACK $\rightarrow$ RST
TCP Connect	nmap -sT 192.168.1.1	15.29 seconds	53/tcp, 80/tcp, 443/tcp	SYN $\rightarrow$ SYN/ACK $\rightarrow$ ACK $\rightarrow$ RST/ACK
UDP Scan	sudo nmap -sU 192.168.1.1	117.28 seconds	53/udp, 67/udp (open filtered)	UDP Probes $\rightarrow$ ICMP Destination Unreachable

2. Technical Analysis & Packet Behavior

A. TCP SYN Scan (-sS) — Half-Open Scanning

- Observed Packet Flow:** Kali sends a TCP packet with the SYN flag set to each target port. When probing open ports (80, 443), the router responds with SYN/ACK. Instead of completing the handshake, Kali immediately terminates the connection by sending an RST packet. Closed ports respond with RST/ACK.
- Speed:** Extremely fast (**1.67 seconds**). Because the scanner drops the connection immediately after receiving SYN/ACK, no socket state overhead is maintained by the operating system.
- Stealth: High.** Since the full TCP 3-way handshake is never completed, application-layer daemons rarely log the connection attempt, allowing the scan to bypass basic application logging.

B. TCP Connect Scan (-sT) — Full Handshake Scanning

- **Observed Packet Flow:** Kali utilizes the operating system's standard connect() system call. For open ports, Kali completes the full 3-way handshake before immediately issuing an RST/ACK to close the socket. Closed ports return a standard RST/ACK.
- **Speed:** Moderate (**15.29 seconds**, is a little bit slower than SYN). Completing the full network connection adds substantial processing time and system overhead for each open port.
- **Stealth: Low.** Because full connections are established, target services register these events. Web servers log connections, and Intrusion Detection Systems (IDS) can easily associate the established sockets with scanning activity.

C. UDP Scan (-sU) — Connectionless Scanning

- **Observed Packet Flow:** Kali transmits raw UDP packets to target ports. For closed ports, the router returns an ICMP Destination Unreachable message.
- **Speed:** Extremely slow (**117.28 seconds**, is it more slower than SYN). UDP scanning is throttled by host operating system rate-limiting on ICMP error messages.
- **Stealth: Moderate.** While UDP lacks TCP handshakes, the high volume of outbound UDP traffic combined with a continuous stream of incoming ICMP Port Unreachable error packets generates a distinct network signature that network monitoring tools easily flag.

4. NSE Scripting Scavenger Hunt

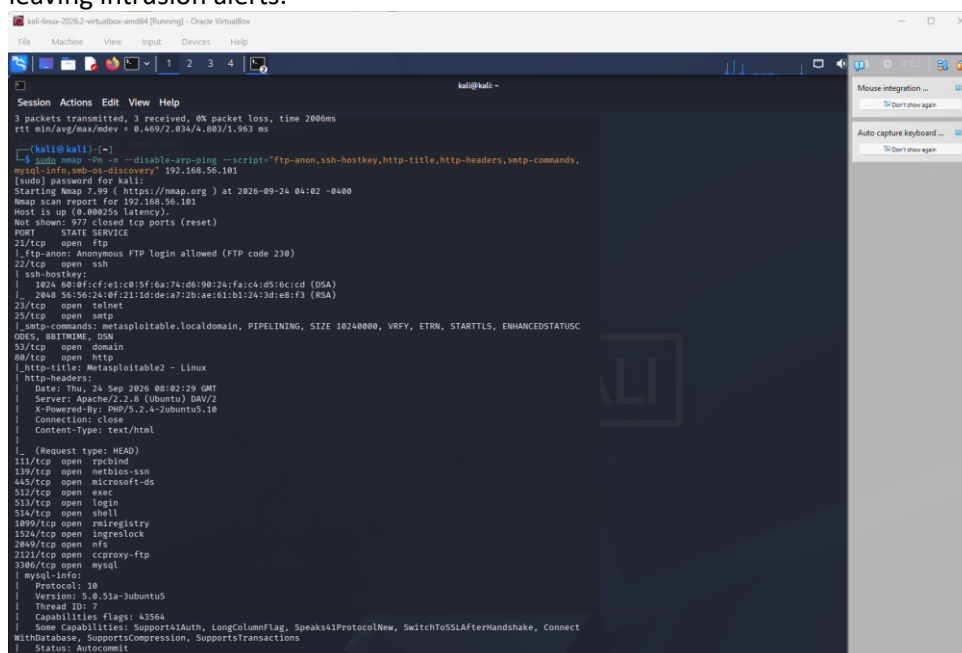
Introduces the Nmap Scripting Engine (NSE) through guided discovery.

- Assign each student or group a different NSE script category (vuln, discovery, safe, auth)
- Run **nmap --script=<category> <target>** against Metasploitable and record findings

Deliverable: A 5-minute lightning talk on what their script category revealed.

Safe: `sudo nmap -Pn -n --disable-arp-ping --script="ftp-anon,ssh-hostkey,http-title,http-headers,smtp-commands,mysql-info,smb-os-discovery" 192.168.56.101`

What it does: Safely grabs banners, web server titles, and OS metadata without risking service crashes or leaving intrusion alerts.



```
kali@kali:~$ sudo nmap -Pn -n --disable-arp-ping --script="ftp-anon,ssh-hostkey,http-title,http-headers,smtp-commands,mysql-info,smb-os-discovery" 192.168.56.101
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-24 04:02 -0400
Nmap scan report for 192.168.56.101
Host is up (0.00025s latency).
Not shown: 972 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
|_ftp-anon Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh
|_ssh-hostkey:
|_  1024 60:8f:cf:fe:c8:5f:6a:74:d6:98:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet
25/tcp    open  smtp
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUS
53/tcp    open  domain
80/tcp    open  http
|_http-title: Metasploitable2 - Linux
|_http-headers:
|_  Date: Thu, 24 Sep 2026 08:02:29 GMT
|_  Server: Apache/2.2.8 (Ubuntu) DAV/2
|_  X-Powered-By: PHP/5.2.4-2ubuntu5.10
|_  Connection: close
|_  Content-Type: text/html
|_ (Request type: HEAD)
111/tcp   open  rsh
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1009/tcp  open  nmap-registry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  cproxy-ftp
3306/tcp  open  mysql
|_mysql-info:
|_  Protocol: 10
|_  Version: 5.0.51a-3ubuntu5
|_  Thread ID: 7
|_  Capabilities Flags: 43564
|_  Some Capabilities: Support4Auth, LongColumnFlag, Speaks41ProtocolNew, SwitchToSSLAfterHandshake, Connect
|_  WithDatabase, SupportCompression, SupportsTransactions
|_  Status: Autocommit
```

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali:~$ sudo nmap -Pn -n --disable-arp-ping --script='smb-enum-shares, rpcinfo, nbstat, dns-nsid, ajp-headers' 192.168.56.101

Nmap scan report for 192.168.56.101
Host is up (0.0018s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
| dns-nsid:
|_ bind.version: 9.4.2
80/tcp    open  http
111/tcp    open  rpcbind
| rpcinfo:
|_ program version  port/proto  service
| 100000 2          111/tcp    rpcbind
| 100000 2          111/udp    rpcbind
| 100003 2,3,4      2049/tcp   nfs
| 100003 2,3,4      2049/udp   nfs
| 100005 1,2,3      48267/udp  mountd
| 100005 1,2,3      51087/tcp  mountd
| 100021 1,3,4      33117/tcp  nlockmgr
| 100021 1,3,4      44840/udp  nlockmgr
| 100024 1          38836/tcp  status
| 100024 1          49002/udp  status
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
512/tcp    open  exec
513/tcp    open  login
514/tcp    open  shell
1099/tcp   open  rmiregistry
1524/tcp   open  ingreslock
2049/tcp   open  nfs
2121/tcp   open  ccproxy-ftp
3306/tcp   open  mysql
5432/tcp   open  postgresql
5900/tcp   open  vnc
6000/tcp   open  X11
6667/tcp   open  irc
8009/tcp   open  ajp13
| ajp-headers:
|_ Content-Type: text/html;charset=ISO-8859-1
8180/tcp   open  unknown
MAC Address: 08:00:27:DD:4F:D0 (Oracle VirtualBox virtual NIC)

Host script results:
| smb-os-discovery:
|_ OS: Unix (Samba 3.0.20-Debian)
|_ Computer name: metasploitable
|_ NetBIOS computer name:
|_ Domain name: localdomain
|_ FQDN: metasploitable.localdomain
|_ System time: 2026-09-24T04:02:29-04:00
Nmap done: 1 IP address (1 host up) scanned in 0.66 seconds

kali@kali:~$
```

Discovery: `sudo nmap -Pn -n --disable-arp-ping --script="smb-enum-shares, rpcinfo, nbstat, dns-nsid, ajp-headers" 192.168.56.101`

What it does: Enumerates SMB share drives, NetBIOS hostnames, and RPC service bindings to map network structures.

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali:~$ sudo nmap -Pn -n --disable-arp-ping --script='smb-enum-shares, rpcinfo, nbstat, dns-nsid, ajp-headers' 192.168.56.101

Nmap scan report for 192.168.56.101
Host is up (0.0018s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
| dns-nsid:
|_ bind.version: 9.4.2
80/tcp    open  http
111/tcp    open  rpcbind
| rpcinfo:
|_ program version  port/proto  service
| 100000 2          111/tcp    rpcbind
| 100000 2          111/udp    rpcbind
| 100003 2,3,4      2049/tcp   nfs
| 100003 2,3,4      2049/udp   nfs
| 100005 1,2,3      48267/udp  mountd
| 100005 1,2,3      51087/tcp  mountd
| 100021 1,3,4      33117/tcp  nlockmgr
| 100021 1,3,4      44840/udp  nlockmgr
| 100024 1          38836/tcp  status
| 100024 1          49002/udp  status
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
512/tcp    open  exec
513/tcp    open  login
514/tcp    open  shell
1099/tcp   open  rmiregistry
1524/tcp   open  ingreslock
2049/tcp   open  nfs
2121/tcp   open  ccproxy-ftp
3306/tcp   open  mysql
5432/tcp   open  postgresql
5900/tcp   open  vnc
6000/tcp   open  X11
6667/tcp   open  irc
8009/tcp   open  ajp13
| ajp-headers:
|_ Content-Type: text/html;charset=ISO-8859-1
8180/tcp   open  unknown
MAC Address: 08:00:27:DD:4F:D0 (Oracle VirtualBox virtual NIC)

Host script results:
| smb-os-discovery:
|_ OS: Unix (Samba 3.0.20-Debian)
|_ Computer name: metasploitable
|_ NetBIOS computer name:
|_ Domain name: localdomain
|_ FQDN: metasploitable.localdomain
|_ System time: 2026-09-24T04:02:29-04:00
Nmap done: 1 IP address (1 host up) scanned in 0.66 seconds

kali@kali:~$
```

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali ~
Session Actions Edit View Help
MAC Address: 08:00:27:DD:4F:D0 (Oracle VirtualBox virtual NIC)

Host script results:
smb-enum-shares:
  account_used: <blank>
  \\192.168.56.101\ADMIN$:
    Type: STYPE_IPC
    Comment: IPC Service (metasploitable server (Samba 3.0.20-Debian))
    Users: 1
    Max Users: <unlimited>
    Path: C:\tmp
    Anonymous access: <none>
  \\192.168.56.101\IPC$:
    Type: STYPE_IPC
    Comment: IPC Service (metasploitable server (Samba 3.0.20-Debian))
    Users: 1
    Max Users: <unlimited>
    Path: C:\tmp
    Anonymous access: READ/WRITE
  \\192.168.56.101\opt:
    Type: STYPE_DISKTREE
    Comment:
    Users: 1
    Max Users: <unlimited>
    Path: C:\tmp
    Anonymous access: <none>
  \\192.168.56.101\print$:
    Type: STYPE_DISKTREE
    Comment: Printer Drivers
    Users: 1
    Max Users: <unlimited>
    Path: C:\var\lib\samba\printers
    Anonymous access: <none>
  \\192.168.56.101\tmp:
    Type: STYPE_DISKTREE
    Comment: oh noes!
    Users: 1
    Max Users: <unlimited>
    Path: C:\tmp
    Anonymous access: READ/WRITE
  nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
  Names:
    METASPLOITABLE<00>  Flags: <unique><active>
    METASPLOITABLE<03>  Flags: <unique><active>
    METASPLOITABLE<20>  Flags: <unique><active>
    WORKGROUP<00>       Flags: <group><active>
    WORKGROUP<1e>       Flags: <group><active>

Nmap done: 1 IP address (1 host up) scanned in 8.47 seconds

kali@kali ~$
```

Auth: `sudo nmap -Pn -n --disable-arp-ping --script="ftp-anon,mysql-empty-password,vnc-info,x11-access" 192.168.56.101`

What it does: Tests for default or missing passwords (such as unauthenticated VNC access or an empty MySQL root password).

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali ~
Session Actions Edit View Help
Names:
  METASPLOITABLE<00>  Flags: <unique><active>
  METASPLOITABLE<03>  Flags: <unique><active>
  METASPLOITABLE<20>  Flags: <unique><active>
  WORKGROUP<00>       Flags: <group><active>
  WORKGROUP<1e>       Flags: <group><active>

Nmap done: 1 IP address (1 host up) scanned in 8.47 seconds

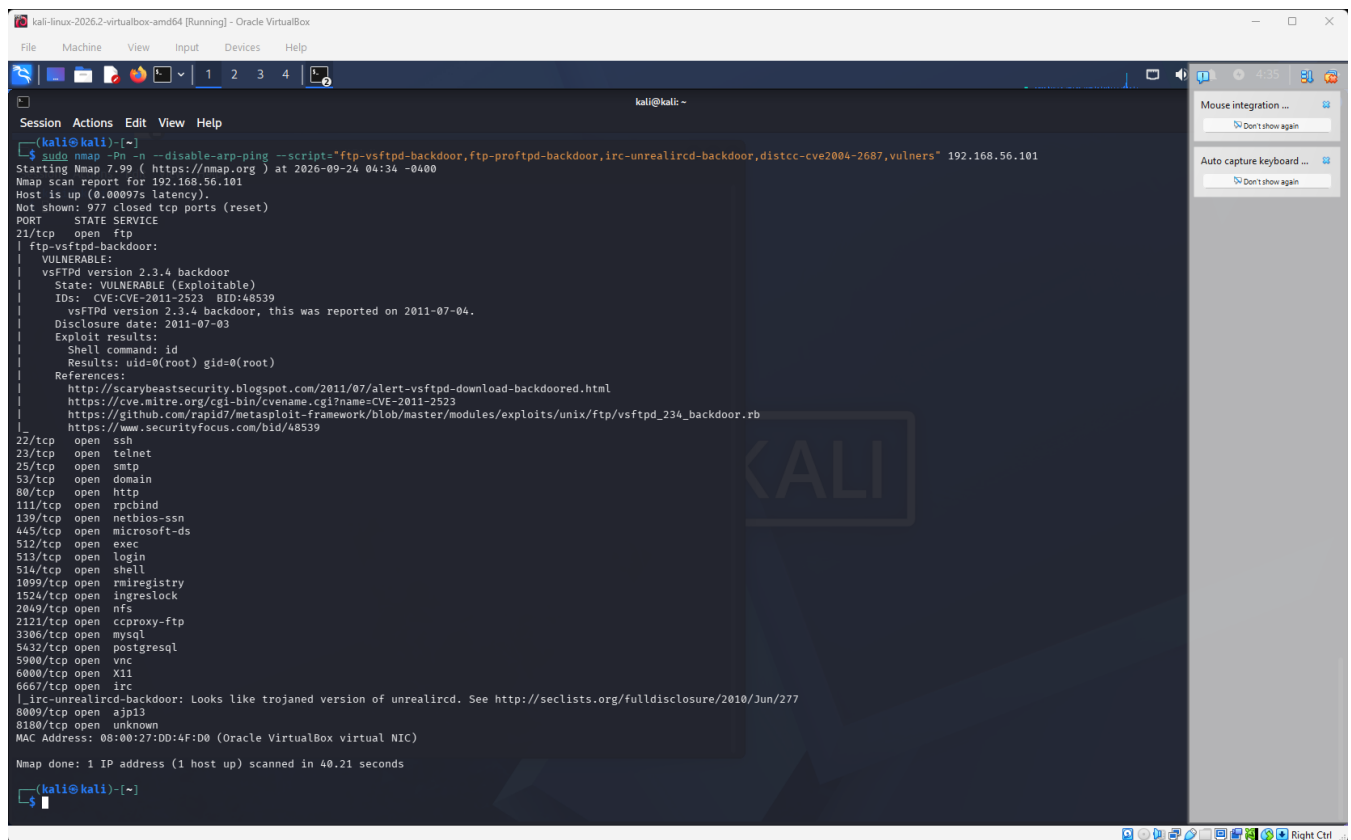
kali@kali ~$ sudo nmap -Pn -n --disable-arp-ping --script="ftp-anon,mysql-empty-password,vnc-info,x11-access" 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-24 04:33 -0400
Nmap scan report for 192.168.56.101
Host is up (0.00072s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
|_mysql-empty-password:
|_  root account has empty password
5432/tcp  open  postgresql
5900/tcp  open  vnc
|_vnc-info:
|_  Protocol version: 3.3
|_  Security types:
|_  VNC Authentication (2)
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:DD:4F:D0 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.28 seconds

kali@kali ~$
```

Vuln Scan: `sudo nmap -Pn -n --disable-arp-ping --script="ftp-vsftpd-backdoor,ftp-proftpd-backdoor,irc-unrealircd-backdoor,distcc-cve2004-2687,vulners" 192.168.56.101`

What it does: Directs checks against Metasploitable's famous backdoors (vsftpd, UnrealIRCd, distcc) to prove remote code execution risks.



5. TryHackMe / HackTheBox Guided Room

Students complete a beginner Nmap-focused room on a legal, structured training platform.

- TryHackMe's "Nmap" room (part of the free "Pre Security" or "Jr Penetration Tester" path) walks through flags step by step
- Students submit screenshots or flag answers as proof of completion

Deliverable: Completed room plus a reflection paragraph on one new flag they learned.

6. Bonus: Capture the Port (CTF-style)

A timed, competitive exercise to reinforce enumeration skills.

- Set up 2–3 Metasploitable/DVWA VMs on an isolated lab network, each with a different open service
- Students race to find and report a hidden "flag" service/port on each machine using only Nmap

Deliverable: First team to correctly report all open ports and services wins.

⚠ A NOTE ON SAFE & LEGAL USE

All of these activities should run inside an isolated lab network or a host-only VM setup (VirtualBox/VMware). Never point Nmap at production networks, other students' devices, or the open internet without explicit written

authorization. Metasploitable and DVWA are free, purpose-built vulnerable VMs designed exactly for this kind of exercise.

Quick Reference: Activities at a Glance

#	Activity	Core Command	Focus Skill
1	Home Network Discovery	<code>nmap -sn</code>	Host discovery
2	Metasploitable Enumeration	<code>nmap -sV -p-</code>	Service & version scanning
3	Scan Type Comparison	<code>-sS / -sT / -sU</code>	Scan technique trade-offs
4	NSE Scavenger Hunt	<code>--script=<category></code>	Scripting engine (NSE)
5	Guided Online Room	Guided walkthrough	Structured self-paced learning
6	Capture the Port (CTF)	Full enumeration	Applied / competitive review